First Year Rossin Engineering
LEHIGH UNIVERSITY

Sensing the World

Project Worksheet

Objective: To break down your project objectives and specifics to make it more efficient to implement. This will help with practicing engineering design.

Submission:

Upload as a group to Gradescope once it has been signed off by course staff.

Team member names:

Anna Zheng, Landon Thomas

Initial Planning

Consider the big picture and then answer the following questions.

1. What is the big system concept your prototype will emulate?

It will emulate a landslide alert system that will trigger a wall.

2. How does this solve a humanitarian problem?

This will solve a humanitarian problem by alerting people of possible landslides in the area, allowing people time to evacuate.

3. What aspects of the big system concept will be omitted in your prototype?

The full alarm system which will be replicated with LEDs.
The actual wall system that will be used.

4. What sensors will be used in this prototype? How are they being used?

Rain Sensor – will be used to detect high water levels in soil.

Light Sensor – will be used to detect the landslide occuring when the soil covers the sensor.

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Project Requirements Document
Use the example from Dr. Valvano covered in class as a guide to complete this section to define your problem.

1. **Project Overview**
    1. *Objectives*:
    Design a alert system with 2 sensors and one alarm. If rain sensor detect high water level in soil and the light sensor is covered by soil, the servo motor arm will push the wall forward and the alarm will be triggered Alarm is the led.
    2. *Process*:
    We will make the first prototype by using a bread board, light sensor, Rain sensor, arduino, LED, and motor. Each part will be tested Separately with the Rain Sensor LED Setup and light Sensor and motor. They will be combining the final prototype.
        1. *Roles and Responsibilities*:
        Landon will develop the hardware and Anna will develop the software. Both will test.
        2. *Interactions with Existing Systems*:
        We will use a launchpad. It will be powered via USB cable with outlet. The components will be selected from lab material.
        3. *Terminology*:
        The light detector Senses light and when light is reduced a signal will be Sent.
        Rain Sensor when wet reduces the voltage which can be used to detect water.
        4. *Security*:
        Wires will have to be covered so they will not get wet
3. **Function Description**
    1. *Functionality*:
    The rain sensor will be in soil When it reaches a certain point it will trigger an alarm of a potential landslide. When the Soil begins to fall it will cover the light Sensor causing a wall to come up and stop landslide.
    2. *Scope*:
    The Scope of the design is to make a prototype to See if project is feasible to be Scaled up.

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3. *Prototypes*:

   We will use a breadboard and launchpad

4. *Performance*:

   If both sensors are triggered, the alarm should set off.
   The response time between the triggered state and alarm signal should be instantaneous.

5. *Usability*:

   The wall can be triggered by a button by an user.

6. *Safety*:

   Making sure no one is near the wall when it is being activated

4. **Deliverables**

   1. *Reports*:

      video will be recorded and placed in the presentation.
      There will also be a demo and reflection.

   2. *Audits*: None

   3. *Outcomes*:

      We will create a prototype system.